Synoptic determination of living/dead and active/dormant bacterial fractions in marine sediments

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Abstract

The most widely used methods for the estimation of the living/dead fractions of bacterial cells involve specific stains that are able to reveal membrane integrity. Here, we have compared two different probes (propidium iodide and ethidium homodimer-2) that have different molecular weights and steric hindrance effects. We have also combined this method with the staining/destaining procedure that is currently used in the identification of potentially active cells. The procedure for marine sediments described here allows the synoptic (i.e. from the same filter) identification of: (i) the number of living bacteria; (ii) the number of active vs. dormant cells within this living fraction; (iii) the bacterial fraction with an intact nucleoid region without membrane integrity; and (iv) dead cells (devoid of the nucleoid region and without membrane integrity). Our results demonstrate that the concentration of propidium is crucial for the correct estimation of the dead bacterial fraction, ethidium homodimer-2 allows efficient and accurate estimates that are independent of the concentrations used and the sample storage. The active bacterial fraction represented *c.* 40% of the total bacterial abundance, the inactive/dormant fraction *c.* 30%, and the dead fraction was, on average, *c.* 30%. This method allows the processing of a large number of samples with high precision and at relatively low cost, and thus it provides additional synoptic insights into the metabolic state of bacteria in marine sediments.

Introduction

The evaluation of the metabolic state of bacteria in marine environments is a key issue in the understanding of their ecological role and their contribution to oceanographic processes. However, it is universally recognised that only a portion of aquatic bacteria are active (Rodriguez *et al.*, 1992; Amann *et al.*, 1995; del Giorgio *et al.*, 1997). An estimation of the metabolically active fraction of marine bacteria has not yet been completely established, especially in marine sediments. In the last years, a number of methodologies have been proposed, such as the demonstration of respiratory electron transport by use of a redox dye (Rodriguez *et al.*, 1992; Proctor & Souza, 2001), the incorporation of radiolabelled substrates (Carman, 1993), direct viable counting (Joux & Lebaron, 1997) and the detection of nucleoid-containing cells by staining (Luna *et al.*, 2002), but all of these have advantages and disadvantages, depending also on the compartment in which they are applied (e.g. water vs. sediments). Among these methods, one of the most widely used to discriminate between active and inactive cells is based on the identification of a visible nucleoid region inside the bacterial cell (Zweifel & Hagström, 1995). This approach assumes that the cells without the nucleoid region are dead or inactive (Zweifel & Hagström, 1995; Choi *et al.*, 1996; Luna *et al.*, 2002).

Estimating the number of dead or inactive bacteria in the sediment is important because sediments contain the largest fraction of the total world prokaryote biomass, and because recent studies provided evidence that a large fraction of the bacteria in different aquatic systems are dead (del Giorgio *et al.*, 1997; Lebaron *et al.*, 2001; Luna *et al.*, 2002;). The most commonly used method for discriminating between living and dead bacteria in environmental samples is based on the use of fluorescent stains. Among these stains, the most frequently used are propidium iodide (PI), ethidium bromide, ethidium monazide, Calcafluor white, SITOX Green and Live/Dead Bac-Light (Lloyd & Hayes, 1995;

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McFeters *et al*., 1995; Porter *et al*., 1996; Lebaron *et al*., 1998; Williams *et al*., 1998; Boulos *et al*., 1999; Sherr *et al*., 1999; Luna *et al*., 2002). Due to their large molecular sizes and shapes, these can have steric hindrance effects, as they can penetrate only those bacterial cells in which membrane integrity is compromised (McFeters *et al*., 1995). The evidence of membrane damage (i.e. a lack of membrane integrity) is a widely accepted criterion for identifying dead cells, which are typically discriminated from living cells by simultaneous staining with a permeant DNA-binding dye (e.g. SYBR Green; Molecular Probes, Milan, Italy). However, results based on membrane integrity should be interpreted with caution. Differential permeabilities of intact membranes of different bacterial strains present limitations to this approach (Gregori *et al*., 2001). This problem is particularly evident also in bacterial cells that are competent for natural transformation, which can display higher membrane permeabilities under the influence of particular environmental conditions (Lorenz & Wackernagel, 1994). This method is also dependent on the properties of each specific stain (e.g. molecular weight and potential for steric hindrance during membrane diffusion), which can lead to bias due to false staining (Williams *et al*., 1998). All these considerations highlight the need for a careful validation of any procedure in order to avoid erroneous interpretations. Moreover, recent studies have indicated that the discrimination between living and dead bacteria is not sufficient to understand whether a living fraction is active or not (Luna *et al*., 2002). The need for new methodologies that combine the enumeration of living/dead cells and of metabolically active cells is therefore apparent.

In order to improve current methodology for estimating the proportions of living and dead cells in marine sediments, we have combined a living/dead counting method that is based on an assessment of membrane integrity with one involving a staining/destaining that is commonly used for the counting of potentially active marine bacteria.

To validate the method for living/dead cell enumeration, we have compared two stains: propidium iodide (the most commonly utilised stain) and ethidium homodimer-2 (EthD-2), which, having a much higher (i.e. double) molecular weight, is a good candidate to overcome problems encountered with the use of propidium iodide. We have used these two stains together with SYBR Green I, which stains all of the bacterial cells independent of cell membrane integrity.

In this study we present a procedure which allows quantitative estimates of: (i) the number of living bacteria; (ii) the number of active and dormant cells in the living fraction; (iii) the fraction of cells with an intact nucleoid region, but lacking membrane integrity; and (iv) the number of dead cells (those lacking a nucleoid region and cell membrane integrity).

Materials and methods

Sediment sampling and processing

Surface sediment samples were collected using a USNEL-type box-corer in the central Adriatic Sea (Italy). The station was located inside Ancona harbour (43°37′320″N, 13°29′971″E, 12.4 m depth) and was characterized by muddy sediments (sand fraction 30.6%; silt fraction 40.9%; clay fraction 28.5%; water content 47.0%). The *in situ* temperature was 10 °C. Fresh subsamples ($n = 3$) were incubated for 15 min with sodium pyrophosphate (at a final concentration of 5 mM) and then sonicated three times (60 W for 1 min; Branson Sonifier 2200, Milan, Italy) for detaching the cells from the sediment particle. This step is necessary for the sedimentary matrix, but could cause damage to cell membrane integrity. In order to exclude any bias due to extraction procedure we tested the effect of ultrasound treatment on the permeability of the membranes on fresh sediments. In addition all staining and sonication procedures were tested on cultured *Vibrio mediterranei* (grown aerobically in marine broth upon exponential-phase, measured as 600 nm absorbance, and then resuspended in 0.9% NaCl solution) and on two different sediment types (one coastal and one deep-sea sediment sample) in order to test the efficiency of the technique utilised.

Prior to staining, the subsamples (1 mL each) were diluted 500-fold and poured onto a 0.2 μm pore-diameter aluminium oxide Anodisc filter under low vacuum (<100 mmHg), and then immediately processed for enumeration of living and dead bacteria by the two double-staining methods. These used SYBR Green I as a permeant stain, and either propidium iodide (PI) or ethidium homodimer-2 (EthD-2) as impermeant stains (Molecular Probes). All counts carried out under epifluorescence microscopy were performed on at least 10 fields per filter and, in any case, until the counting of 400 cells. Bacterial counts were referred to as mL of sediment.

Enumeration of living/dead bacterial fractions with the fluorescent probes

Living and dead bacterial cells were counted under a double-staining method that used SYBR Green I and two stains that were assumed to penetrate only into compromised cells (PI and EthD-2). The cells stained with PI or EthD-2 were distinguished by their red fluorescence from the cells stained yellow-green by SYBR Green I. The sediment samples were treated with 10 μL of SYBR Green I solution (Molecular Probes) diluted 1000-fold in Milli-Q water (Millipore, Milan, Italy), and with 10 μL PI or EthD-2 solution containing 1, 5, 10, 20, 30 and 40μg mL^{-1} of either stain. The diluted PI and EthD-2 solutions were prepared from their respective stock solutions. After staining (20 min in the dark), the

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sediment-bearing filters were mounted on microscope slides with 50% glycerol and 50% phosphate-buffered saline with 0.5% ascorbic acid (10 μL under and 10 μL over the filter). The stained filters were examined by epifluorescence microscopy (magnification ×1000; Zeiss Axioskop 2, Zeiss, Milan, Italy) using the BP 490 filter set for SYBR Green I (yellow-green florescence) and the BP 545 filter for PI and EthD-2 (red fluorescence).

The two different fluorescent stains used (PI and EthD-2) are nucleic acid-specific ('dead cell' stains; Shumann *et al.*, 2003), and they were applied to determine the integrity of the bacterial membrane. Both of these stains are charged and/or have structures that can result in steric hindrance, which can hamper their diffusion through the cell membrane. PI is a double-charged phenanthridium derivative and is one of the most common stains for dead cells (Haugland, 2001). EthD-2 consists of two phenanthridinium fluorophores with a large spacer group, and it has four charges. We applied both stains with the permeant SYBR Green I. Under blue excitation, the intact cells appeared green due to SYBR Green I staining and the permeabilised cells appeared red due to the additional or replaced intercalation by PI or EthD-2 in the nucleic acids.

Estimating the active/dormant bacterial fractions and the dead fraction by staining/destaining

Recognition of the bacterial active fraction is based on the presence of a visible nucleoid region (Zweifel & Hagström, 1995). To enumerate the number of nucleoid containing cells (NuCC), we applied a staining/destaining procedure that was adapted for sediment samples by Luna *et al.* (2002). After the living/dead counting (as described above), the sediment-bearing filters were placed in a filtration set-up and rinsed with 10 mL of 2-propanol over 10 min (to remove the excess of stain). This step allows the fraction of stain not bound to the nucleoid to wash away, thus allowing the estimation of the non-nucleoid containing cells (non-NuCC). The NuCC counting was carried out under epifluorescence microscopy (magnification, ×1000; Zeiss Axioskop 2), where the nucleoid of living bacteria appeared green due to the excitation of the SYBR Green I dye with which the cells have been stained. The same procedure was also used for destaining the red fluorescent cells (i.e. those with compromised membrane integrity) on the filters stained using PI and EthD-2. This allowed a distinction to be made between actual dead cells (without a visible nucleoid region) and the red-stained bacteria with a still visible nucleoid region. After the counting of bacteria under blue excitation (for the NuCC counts), the sediment-bearing filters were immediately examined for the presence of red cells with a visible nucleoid region, and these nucleoid-containing cells were counted.

Effects of storage on estimates of living/dead counts

To estimate the effects of the preservative on the living and dead counts, subsamples of fresh sediment were compared with subsamples (*c.* 0.5 mL) that had been fixed with formaldehyde (final concentration 2%) and processed immediately after fixation (2 h) or after 10, 20, 30 and 40 days. All of the samples were processed as described above for living/dead counting (the final concentration for both PI and EthD-2 was 20 μg mL^{-1}). After fixation all samples were kept in the dark at 4 °C until analysis.

Statistical analyses

The differences between the living/dead counts and the active/dormant bacterial fractions were analysed statistically using ANOVA, followed by the *post-hoc* Newman–Keuls test (NK) to discern significant effects (significance at $P < 0.05$).

Results and discussion

Comparison of propidium iodide and ethidium homodimer-2 in the determination of the living/dead bacterial fractions in sediment samples

The use of nucleic acid-specific permeant and nonpermeant stains under epifluorescence microscopy is the most common method for the counting of living and dead bacteria in environmental samples (Humphreys *et al.*, 1994; Mason *et al.*, 1995; Haugland, 1996; Williams *et al.*, 1998; Shumann *et al.*, 2003), but there is increasing evidence that bias in the discrimination between living and dead bacteria can occur if some methodological aspects are not fully clarified. We assumed that the bacteria without complete membrane integrity were dead or were competent for natural transformation. We also demonstrated that the use of sonication did not alter the results obtained using different staining procedures (Table 1). In fact, the staining with ethidium homodimer-2 (EthD-2) with and without ultrasound treatment resulted in an identical percentage of dead/living cells. Similar results were obtained when we used propidium iodide (PI).

PI has a molecular weight of 668.4 and is thus assumed to be unable to penetrate cell membranes. Evidence increasingly suggests that use of PI can lead to an overestimation of the size of the dead bacterial fraction, and can thus result in a bias between the living/dead bacterial counts (Williams *et al.*, 1998; Barbesti *et al.*, 2000). In order to identify the possible use of alternative stains, we compared the effects of PI and EthD-2 on bacterial cells in both fresh and fixed sediment samples. When the two stains were used at concentrations of 10–20 μg mL^{-1}, the living bacterial counts obtained using SYBR Green I together with PI or EthD-2 in

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Table 1. Effects of ultrasound treatment on counts of total bacteria and their dead and living fraction in marine sediments carried out after staining with propidium iodide (PI) and ethidium homodimer-2 (EthD-2)

		Total counts				Dead cells				Living cells				Dead cells		Living cells	
		PI		EthD-2		PI		EthD-2		PI		EthD-2		PI	EthD-2	PI	EthD-2
Treatment	Conc. ($\mu g\,mL^{-1}$)	Cells × $10^7\,mL^{-1}$	Std	Cells × $10^7\,mL^{-1}$	Std	Cells × $10^7\,mL^{-1}$	Std	Cells × $10^7\,mL^{-1}$	Std	Cells × $10^7\,mL^{-1}$	Std	Cells × $10^7\,mL^{-1}$	Std	%	%	%	%
Not ultrasound	20	3.21	0.43	1.89	0.20	1.14	0.15	0.91	0.09	2.07	0.28	0.98	0.11	36	48	64	52
	40	3.27	0.67	4.54	0.98	2.10	0.43	1.74	0.37	0.17	0.24	2.80	0.61	64	38	36	62
With ultrasound	20	20.92	1.23	24.44	1.15	6.48	0.38	11.73	0.55	14.43	0.85	12.71	0.60	31	48	69	52
	40	24.94	2.56	21.20	2.45	17.96	1.84	9.96	1.15	6.98	0.72	11.24	1.30	72	47	28	53
														ns*	ns*	ns*	ns*

*ns, not significant. Std, standard deviation.

Table 2. Comparison of the staining with propidium iodide (PI) and ethidium homodimer-2 (EthD-2) on counts of total bacteria and their living and dead fraction in marine sediments

	Total counts				Dead cells				Living cells			
	PI		EthD-2		PI		EthD-2		PI		EthD-2	
Concentrations ($\mu g\,mL^{-1}$)	Cells × $10^7 mL^{-1}$	Std	Cells × $10^7 mL^{-1}$	Std	Cells × $10^7 mL^{-1}$	Std	Cells × $10^7 mL^{-1}$	Std	Cells × $10^7 mL^{-1}$	Std	Cells × $10^7 mL^{-1}$	Std
1.0	9.09	2.09	20.58	3.46	2.00	1.24	6.04	0.45	7.09	1.20	14.54	3.01
5.0	10.65	1.18	21.43	0.52	3.22	0.61	6.74	0.43	7.43	1.30	14.69	0.09
10.0	27.36	8.94	19.86	0.70	6.95	1.79	6.21	0.19	20.41	7.31	13.65	0.51
20.0	27.84	5.41	24.93	1.43	8.99	1.60	6.61	0.81	18.85	3.82	18.33	0.63
30.0	26.96	2.11	22.43	6.10	16.08	2.36	6.24	1.20	10.88	0.98	16.19	4.90
40.0	21.77	0.97	23.13	5.07	19.33	1.18	7.69	0.90	2.43	0.33	15.44	4.18

Std, standard deviation.

fresh sediment samples were not significantly different (on average, 19.5 ± 5.6 and $16.0 \pm 0.6 \times 10^7$ cells mL^{-1}, respectively; see Table 2). However, at lower (e.g. 1.0–5.0 $\mu g\,mL^{-1}$) or higher (e.g. 30–40 $\mu g\,mL^{-1}$) concentrations, the living bacterial counts displayed significant differences (NK, $P < 0.001$) between the two stains. Indeed, the counts of the dead (red-stained) cells using PI increased progressively with increasing stain concentration, whereas the counts using EthD-2 did not. Such differences are potentially related to the much higher molecular weight of EthD-2 (1292.71), which will prevent this compound from penetrating bacterial cells that do not have large pores, or that have suffered membrane damage, which will hereafter be described as cells lacking in membrane integrity.

The changes in the proportions of the living and dead bacteria at varying PI and EthD-2 concentrations are shown in Fig. 1. Using PI, the living bacterial fraction ranged from 11% to 80%, with significantly higher values seen at PI concentrations <20 $\mu g\,mL^{-1}$ than at all other concentrations (NK, $P < 0.05$). The living counts were drastically reduced at PI concentrations $>30\,\mu g\,mL^{-1}$, when the dead bacterial count reached 80% of the total bacterial counts (i.e. significantly higher than at any other PI concentration; NK, $P < 0.001$). Conversely, with EthD-2, the living bacterial fraction remained rather constant at all concentrations (in the range of 67–74%, even at very low stain additions). In addition, the cells in the sediment deposit on the filter membranes were more clearly visible and differentially stained, enabling better discrimination of the red-stained cells (Fig. 2). Shumann *et al.* (2003) also compared living/dead cell counts obtained after PI staining. They used a single PI concentration of 3.3 mM (2200 $\mu g\,mL^{-1}$), which was 100-fold greater than that used in all previous studies, to allow a longer staining and fluorescence, thus facilitating the counting (personal observations). They also tested EthD-2 at a concentration of 0.5 μM (0.625 $\mu g\,mL^{-1}$). With PI staining, the counts were two- to fourfold higher compared to EthD-2 staining. Our results suggest that EthD-2 staining yields more reliable estimates of dead bacterial cells and avoids the counting of artefacts of staining.

This was further confirmed by an experiment in which we stained a strain of *Vibrio mediterranei* in exponential growth phase with PI. During exponential growth, the large majority of cells were actively growing and the number of 'dead' cells was negligible, but the use of high PI concentrations (i.e. above 20 $\mu g\,mL^{-1}$) led to an increase of the number of apparently dead cells, which, at 40 $\mu g\,mL^{-1}$, became dominant (i.e. $>50\%$; Fig. 3).

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In addition, this protocol was applied to two different sediment types (one coastal and one deep-sea sediment sample) for which three different PI and EthD-2 concentrations were compared (1, 20 and 40 μg mL^{-1}). These results further confirm our findings obtained from other samples and from the *V. mediterranei* culture (Table 3), indicating that the number of dead cells, both in coastal and deep-sea sediments, increased with increasing PI concentration but not with increasing EthD-2 concentrations.

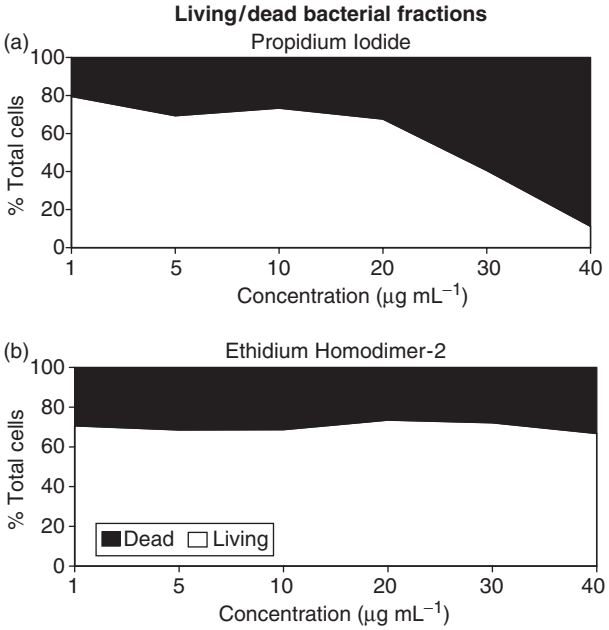

Fig. 1. Percentages of living and dead bacterial fractions obtained after staining with: (a) propidium iodide and (b) ethidium homodimer-2 at different concentrations (1, 5, 10, 20, 30 and 40 μg mL^{-1}).

Effects of sample preservation on living/dead cell counting

The effects of sample preservation on living/dead bacterial counts is illustrated in Fig. 4. Although the living counts at optimal PI concentrations (i.e. 20 μg mL^{-1}) were identical to those for EthD-2 when applied to fresh samples, sample storage conditions had a critical effect on cell counts using PI staining. Indeed, in fixed samples stained with PI, the number of living cells apparently decreased during storage, with a significant decrease already evident after 2 h of fixation (NK, $P < 0.05$). After 40 days of preservation, the dead-cell percentage increased up to *c*. 50% when compared with fresh cells. Conversely, the percentage of living and dead cells in samples stained with EthD-2 did not change significantly throughout the study period (Fig. 4). Williams *et al*. (1998) reported that the use of PI was not effective in water samples preserved in formalin and glutharaldehyde. A negative effect of formaldehyde fixation was also reported by del Giorgio *et al*. (1996). Formalin permeabilizes cell

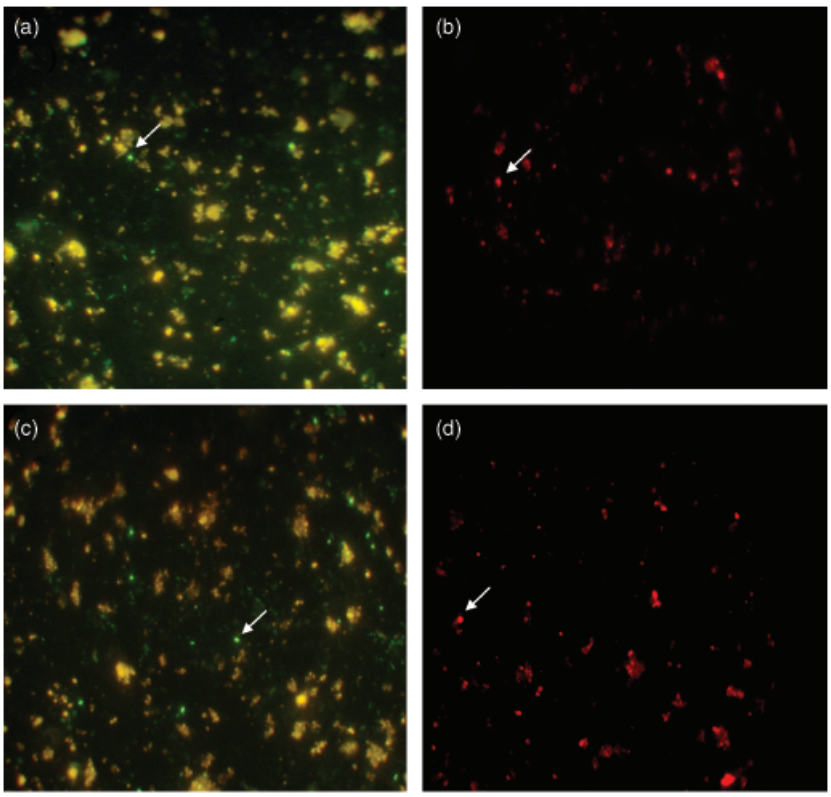

Fig. 2. Marine bacteria examined using epifluorescence microscopy (magnification, ×1000; Zeiss Axioskop 2, BP 490 filter set for SYBR Green I for the yellow-green florescence and the BP 545 filter for propidium iodide and ethidium homodimer-2 for the red fluorescence). Fig. 2a and b illustrate bacteria stained with the green SYBR I (living cells) and bacteria stained with propidium iodide (dead cells), respectively. Fig. c and d illustrate bacteria stained with the green SYBR I (living cells) and bacteria stained with ethidium homodimer-2 (dead cells), respectively. Arrows indicate as it appears the typical stained cell.

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membranes, and consequently allows the penetration of PI into all of the cells regardless of their physiological state. Our results support these findings, suggesting that formaldehyde fixation alters the living/dead counts by increasing the permeability of the bacterial membranes to PI. The use of EthD-2 circumvents this problem, probably due to its greater molecular weight and molecular size and shape. Our results suggest that estimates of the dead bacterial fraction reported in the literature and based on fixed samples might have been overestimated by up to *c*. 50% (for a sampling storage of 40 days), and indicate that future studies should use EthD-2 to avoid any bias related to sample fixation.

Analysis of the viability of bacteria in marine sediment: active vs. dormant cells

When the living/dead staining protocol is coupled to the staining/destaining procedure, it is possible to gather additional synoptic information on the physiological state of benthic bacteria. The fraction of active bacteria can then be estimated as the number of bacteria displaying a visible nucleoid region (the nucleoid containing cells, NuCC), whereas the cells without a visible nucleoid region can be defined as inactive/dormant or dead (Gasol *et al*., 1995).

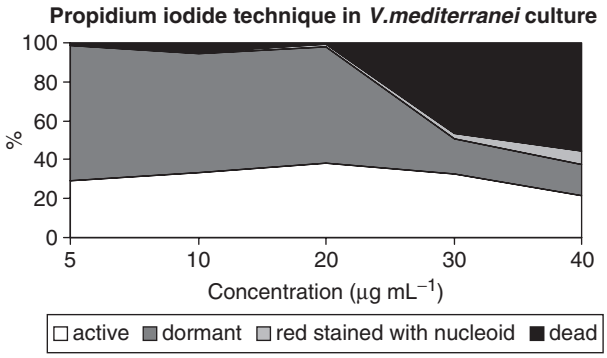

Fig. 3. Percentage of active, dormant, dead and red-stained cells but with nucleoid region, obtained using propidium iodide on a bacteria strain (e.g. *Vibrio mediterranei*) in an exponential growth phase.

When these counts of bacteria with and without a visible nucleoid are combined with the living/dead counts obtained by staining with PI or EthD-2, the following three cell categories can be distinguished: (i) active cells (NuCC counts); (ii) dormant/inactive cells (non-NuCC subtracted from the dead cell count); and (iii) dead cells (red-stained

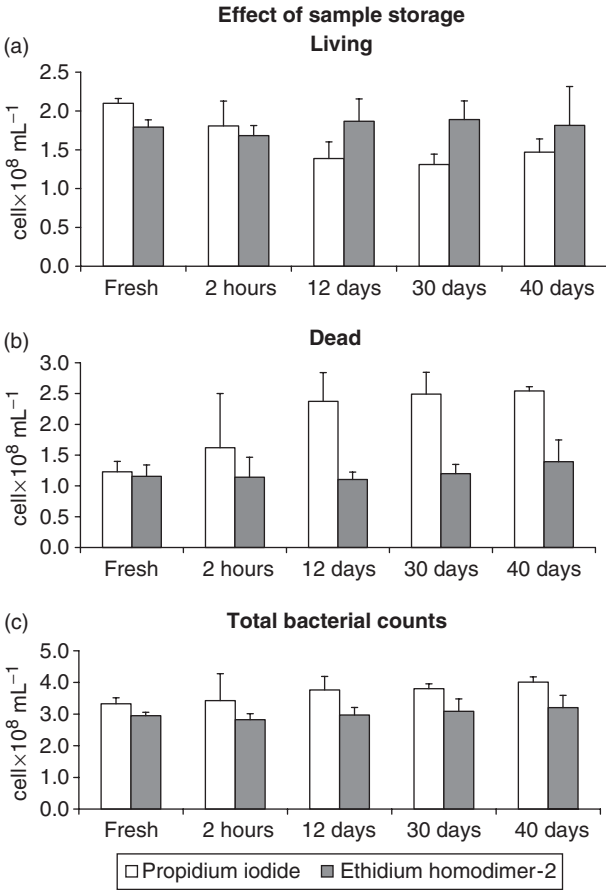

Fig. 4. Number of living (a), dead (b) and total bacterial counts (c), obtained using propidium iodide and EthD-2 in fresh samples and after storage in formaldehyde. Bacterial numbers are given together with standard deviations ($n = 3$).

Table 3. Comparison of the two stains on different sediment types from various locations

		Total counts				Dead cells				Living cells				Dead cells		Living cells	
		PI		EthD-2		PI		EthD-2		PI		EthD-2		PI		EthD-2	
Sediments samples (Location)	Conc. (μg mL^{-1})	Cells × 10^8 mL^{-1}	Std	Cells × 10^8 mL^{-1}	Std	Cells × 10^8 mL^{-1}	Std	Cells × 10^8 mL^{-1}	Std	Cells × 10^8 mL^{-1}	Std	Cells × 10^8 mL^{-1}	Std	%	%	%	%
Coastal sediment 23 m	1	4.23	0.30	4.98	0.49	0.35	0.05	0.85	0.10	3.88	0.26	4.13	0.40	8.3	17.1	91.7	82.9
(Manfredonial	20	5.58	0.57	5.59	0.90	1.02	0.11	1.29	0.13	4.56	0.46	4.30	0.77	18.3	23.0	81.7	77.0
Gulf–Adriatic Sea)	40	4.49	0.76	4.86	0.31	2.70	0.66	1.22	0.12	1.80	0.10	3.63	0.44	60.1	25.2	39.9	74.8
Deep sediment 2850 m	1	1.02	1.16	1.09	0.05	0.13	0.02	0.32	0.15	0.89	0.14	0.76	0.10	12.9	29.7	87.1	70.3
(Western	20	0.99	0.16	1.14	0.12	0.35	0.05	0.42	0.10	0.64	0.11	0.72	0.02	35.5	37.0	64.5	63.0
Mediterranean Sea)	40	1.02	0.08	0.99	0.29	0.66	0.15	0.38	0.11	0.35	0.07	0.61	0.19	65.2	38.3	34.8	61.7

Std, standard deviation.

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bacteria; Luna *et al*., 2002). In addition, the new simultaneous application of the two different protocols described here allows the identification of cells with compromized membrane integrity that still display a visible nucleoid region (Fig. 5).

The percentage of NuCC (active bacteria) varied according to the dye used (from 5–32% with PI, to 30–47% with EthD-2). The NuCC counts obtained by staining with SYBR Green I and EthD-2 were more constant at different EthD-2 concentrations and were always significantly higher (except at 20 μg mL^{-1}, where the counts were identical) than those obtained using PI (ANOVA, $P < 0.05$). The results reported here using SYBR Green I and EthD-2 together are comparable to those reported in most studies on marine and freshwater systems (Gasol *et al*., 1995; Zweifel & Hagström, 1995; Choi *et al*., 1996; Heissenberger *et al*., 1996; Karner & Fuhrman, 1997; Vosjan & van Noort, 1998; Bermann, 2001; Luna *et al*., 2002). The fraction of dormant/inactive bacteria was again more constant in samples stained with EthD-2 (range: 23–37% of the total bacterial number) than in samples stained with PI (range: 7–48% of the total bacterial number). Previous studies have reported similar counts of dormant/inactive bacteria in coastal sediments (Luna *et al*., 2002) and seawater samples (Pinhassi *et al*., 1999).

While a number of studies on the active and dormant bacterial fractions in marine sediments are available for comparison (Proctor & Souza, 2001; Stenström *et al*., 2001), estimates of the dead bacterial fraction are extremely scarce (Luna *et al*., 2002). In the present study the dead fraction determined as red fluorescent cells after EthD-2 staining represented, on average, *c*. 30% of total bacterial counts. However, the combination of the living/dead and the active/dormant methods allowed the identification of a fourth group of cells: the bacteria with a compromised membrane (which should to be dead), but with a visible nucleoid; such cells may retain a vital function in the total population. The presence of this cell category, which comprised *c*. 6% of the total cell abundance (equivalent to *c*. 20% of the red fluorescent cells), could be explained in two ways. Such bacterial cells could be competent in natural genetic transformation, owing to large pores in their membranes that render them able to take up high-molecular-weight compounds (Lorenz & Wackernagel, 1994). Competent cells are indeed able to take up DNA fragments of >1000 bp, equivalent to >300 000 Da, and as such are certainly also permeable to EthD-2 and PI. An alternative explanation is that permeable bacteria are 'moribund' or freshly killed cells, i.e. cells with damaged membranes but with a nucleoid region that is still intact. The combined methodology reported here results in the accurate quantification of active, dormant/inactive and dead cells, with the advantage that it allows synoptic counts of all of these components. In addition, this new procedure identifies the presence of a minor, but not negligible, fraction of cells without an intact membrane but with a visible nucleoid region. Further studies are needed to determine the characteristics and ecological role of this bacterial fraction that is present in marine sediments.

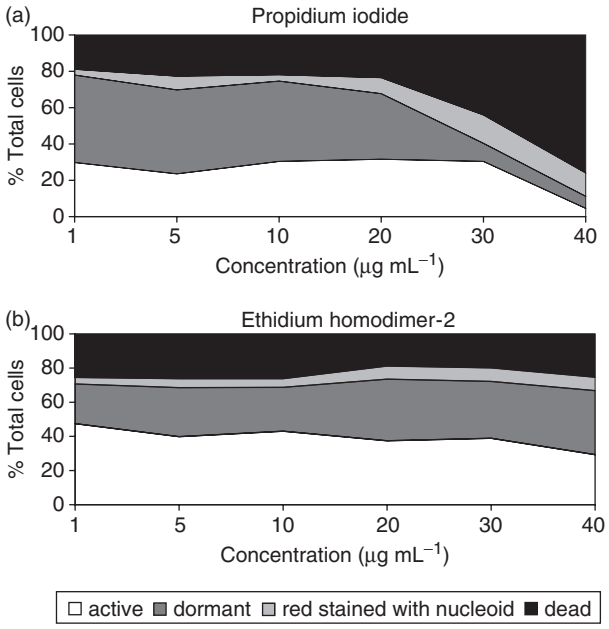

Fig. 5. Effect of different concentrations (1, 5, 10, 20, 30 and 40 μg mL^{-1}) of the two stains propidium iodide (a), and EthD-2, (b) on the percentage of active, dormant, red-stained (with nucleoid region) and dead cells in marine sediments.

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